

AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2018

Physics Paper I

Time: 40 minutes Marks: 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.

Correct Way	Incorrect Ways
1 ☐ A ☐ B ☒ C ☐ D	1 ☐ A ☐ B ☒ C ☐ D
	2 ☐ A ☐ B ☒ C ☐ D
	3 ☐ A ☐ B ☒ C ☐ D
	4 ☐ A ☐ B ☒ C ☐ D

Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The length of a metallic rod is recorded as 8.50×10^5 m.

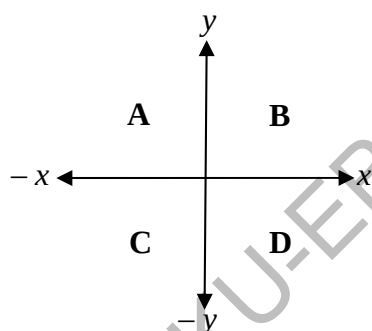
The number of significant figures in the given measurement is

- A. three.
- B. four.
- C. five.
- D. six.

2. Which of the following shows the CORRECT dimensions of velocity, force and momentum?

	Velocity	Force	Momentum
A	LT^{-1}	MLT^{-2}	MLT^{-1}
B	MLT^{-2}	LT^{-1}	LT^2
C	LT^{-1}	LT^{-1}	MLT^{-2}
D	MLT^{-2}	MLT^2	LT

3. R_x and R_y are components of a vector R . If R_x is positive and R_y is negative, then the quadrant in which vector R lies is



4. If the magnitude of a force is 10 N, then the magnitude of its rectangular components can be

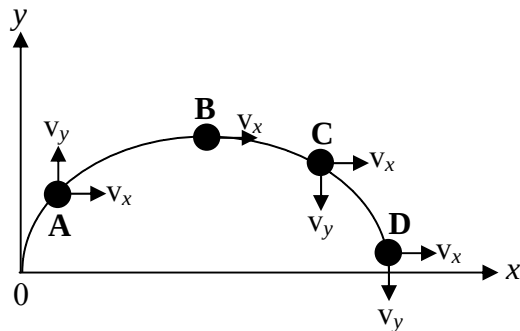
- A. 2 N and 8 N.
- B. 4 N and 7 N.
- C. 5 N and 5 N.
- D. 6 N and 8 N.

5. The vector product of two unit vectors $\hat{j}$ and $\hat{i}$, which are perpendicular to each other, is

- A. 0
- B. 1
- C. $-\hat{k}$
- D. $\hat{k}$

6. The given diagram shows the projectile motion of a ball.

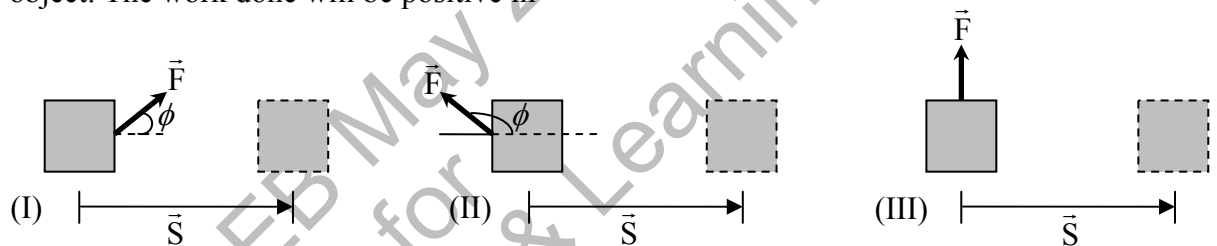
The magnitude of the vertical component of velocity (v_y) will be maximum at point



7. The horizontal range of a projectile is maximum at an angle of

- A. 0°
- B. 30°
- C. 45°
- D. 60°

8. In the given figures, three cases are shown in which a constant force is trying to move an object. The work done will be positive in



- A. I only.
 - B. II only.
 - C. III only.
 - D. I and II.
9. A satellite is moving at a constant speed in a circular orbit due to gravity.

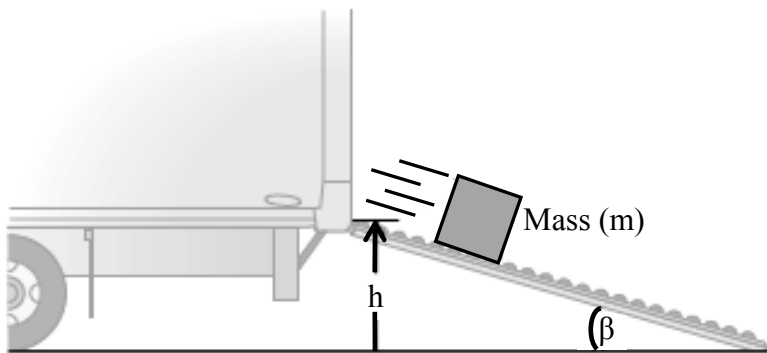


Which of the following statements is CORRECT about the above condition?

- A. The earth does zero work on the satellite.
- B. The earth does positive work on the satellite.
- C. The earth does negative work on the satellite.
- D. The concept of work is not applicable in here.

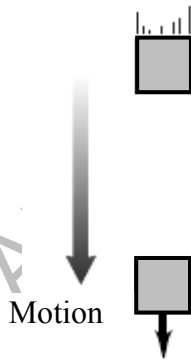
PLEASE TURN OVER THE PAGE

10. A packet of mass (m) is unloaded from a truck by using inclined plane from a height (h) as shown in the given figure.



If friction is negligible, then the kinetic energy on the packet when it reaches the ground will be equal to

- A. $\frac{1}{2} mgh$
 B. mgh
 C. $2mv^2$
 D. mv^2
11. An object experiences a free fall motion as shown in the given figure.

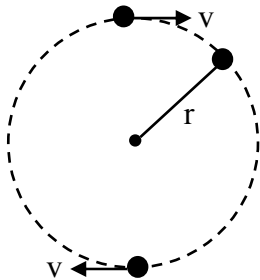


Which of the following is TRUE with respect to the work done (W) by the gravitational force and the object's gravitational potential energy (U)?

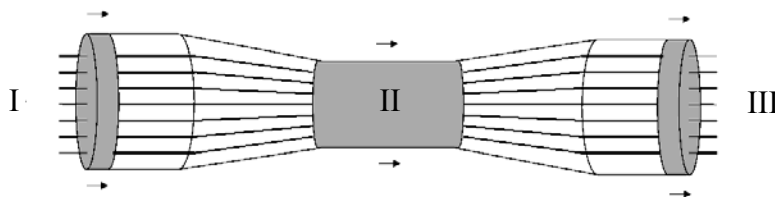
	W	U
A	Positive	Decreases
B	Negative	Decreases
C	Negative	Increases
D	Positive	Increases

12. The velocity which keeps a satellite in its orbit is known as
- A. escape velocity.
 B. critical velocity.
 C. angular velocity.
 D. artificial velocity.

13. In the given diagram, an object is rotating with a speed which is increasing. The angular acceleration of the object is in the same direction as its

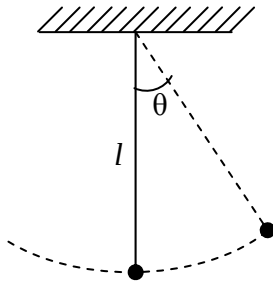


- A. angular velocity.
 - B. tangential velocity.
 - C. linear displacement.
 - D. radial displacement.
14. The angle subtended at the centre of a circle by a particle moving along its circumference in a given time is called angular
- A. velocity.
 - B. momentum.
 - C. acceleration.
 - D. displacement.
15. Which of the following is used to obtain a smooth motion?
- A. Ripples
 - B. Upthrust
 - C. Turbulent
 - D. Streamline
16. At which cross sectional area(s) of the given pipe, the water flow is fastest?



- A. I only
- B. II only
- C. III only
- D. I and III

17. The time period of a simple pendulum is independent of



- I. mass (m) of the bob
 - II. length (l) of a pendulum
 - III. acceleration due to gravity (g)
- A. I only.
 - B. II only.
 - C. I and III.
 - D. II and III.
18. If a body is executing simple harmonic motion (SHM), then the total energy of the body is directly proportional to the
- A. amplitude.
 - B. square of the amplitude.
 - C. reciprocal of the amplitude.
 - D. square root of the amplitude.
19. Resonance is a phenomenon in which a vibrating system or external force drives another system to oscillate with greater amplitude at specific frequencies.
- This phenomenon helps in determining the
- A. forced vibration.
 - B. natural frequency.
 - C. energy dissipation.
 - D. amplitude of a vibrating body.
20. Which of the following would undergo damped oscillation?
- A. Prongs of a tuning fork
 - B. Shock absorber of a car
 - C. Bob of a simple pendulum
 - D. Vibrating wire of a sonometer

21. Which of the following factors affects the speed of sound in air?
- I. Density
 - II. Pressure
 - III. Temperature
- A. I and II
 - B. I and III
 - C. II and III
 - D. I, II and III
22. In the phenomenon of Doppler's effect, when a source of sound moves towards a stationary listener, it results in a decrease of the
- A. pitch of sound.
 - B. velocity of sound.
 - C. frequency of sound.
 - D. wavelength of sound.
23. Which phenomena show the wave nature of light?
- I. Refraction
 - II. Interference
 - III. Compton effect
 - IV. Photo electric effect
- A. I and II
 - B. I and IV
 - C. II and III
 - D. III and IV
24. According to Huygen's principle, light travels in the form of
- I. photons
 - II. corpuscles
 - III. wavefronts
- A. I only.
 - B. III only.
 - C. I and II.
 - D. II and III.
25. When a visible light ray passes through a small opening of a narrow slit, the longer wavelength, as compared to shorter wavelength, of the light ray will
- A. reflect less.
 - B. diffract less.
 - C. reflect more.
 - D. diffract more.

26. According to Boyle's law, pressure is inversely proportional to volume of a gas at constant
- I. mass
 - II. density
 - III. temperature
- A. I only.
 - B. III only.
 - C. I and II.
 - D. II and III.
27. A thermodynamics process in which an ice cube starts melting and converts into water droplets while keeping the temperature of its surrounding constant is called an
- A. isobaric process.
 - B. isochoric process.
 - C. adiabatic process.
 - D. isothermal process.
28. According to the 2nd law of thermodynamics, heat can be converted into mechanical work if the system contains
- A. two heat reservoirs at the same temperature.
 - B. two heat reservoirs at different temperatures.
 - C. an engine and a heat reservoir at the same temperature.
 - D. an engine and two heat reservoirs at different temperatures.
29. A Carnot heat engine is an engine that operates on the reversible Carnot cycle.
- Its efficiency depends upon the temperature of
- A. the surrounding.
 - B. the hot reservoir only.
 - C. the cold reservoir only.
 - D. both hot and cold reservoirs.
30. Entropy is best described as the
- A. decrease in energy in the system.
 - B. increase in energy in the surrounding.
 - C. unavailability of energy in the system.
 - D. degradation of energy in the environment.

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